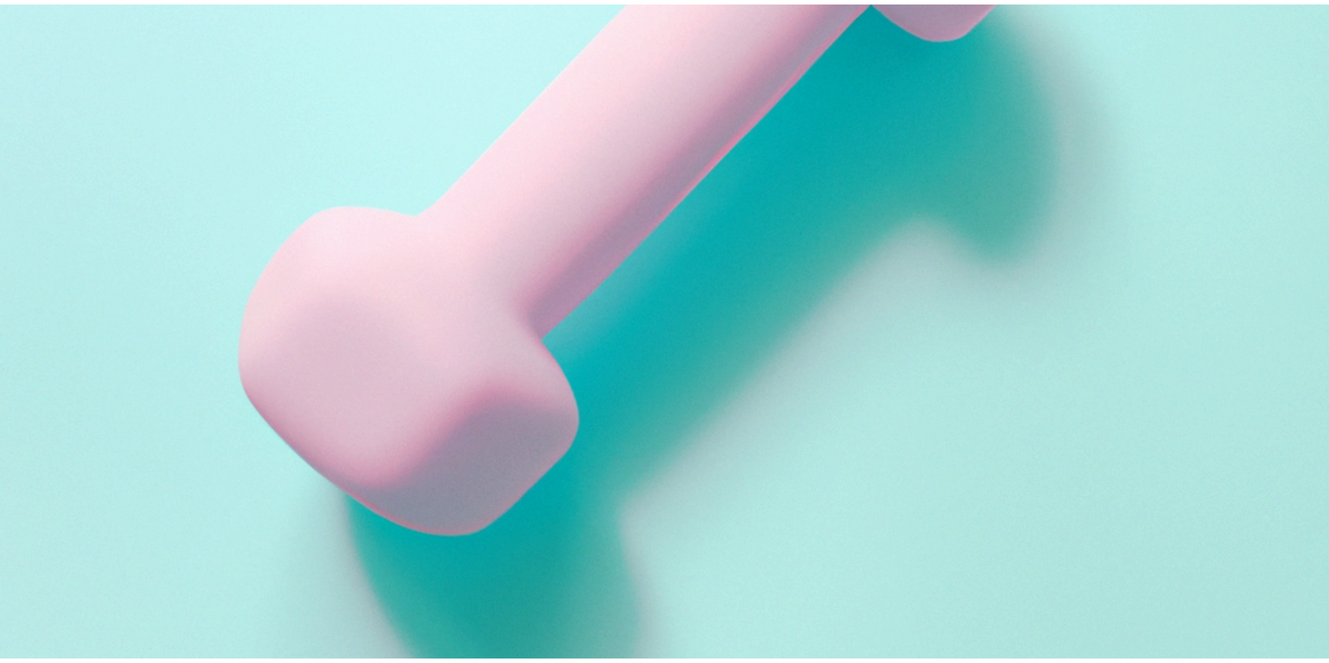


You are a product manager for a fitness studio and are interested in understanding the current demand for digital fitness classes. You plan to conduct a market analysis in Python to gauge demand and identify potential areas for growth of digital products and services.

The Data

You are provided with a number of CSV files in the "Files/data" folder, which offer international and national-level data on Google Trends keyword searches related to fitness and related products.

workout.csv

Column	Description
`'month'`	Month when the data was measured.
`'workout_worldwide'`	Index representing the popularity of the keyword 'workout', on a scale of 0 to 100.

three_keywords.csv

Column	Description
`'month'`	Month when the data was measured.
`'home_workout_worldwide'`	Index representing the popularity of the keyword 'home workout', on a scale of 0 to 100.
`'gym_workout_worldwide'`	Index representing the popularity of the keyword 'gym workout', on a scale of 0 to 100.
`'home_gym_worldwide'`	Index representing the popularity of the keyword 'home gym', on a scale of 0 to 100.

workout_geo.csv

Column	Description
`'country'`	Country where the data was measured.
`'workout_2018_2023'`	Index representing the popularity of the keyword 'workout' during the 5 year period.

three_keywords_geo.csv

Column	Description
`'country'`	Country where the data was measured.
`'home_workout_2018_2023'`	Index representing the popularity of the keyword 'home workout' during the 5 year period.
`'gym_workout_2018_2023'`	Index representing the popularity of the keyword 'gym workout' during the 5 year period.
`'home_gym_2018_2023'`	Index representing the popularity of the keyword 'home gym' during the 5 year period.

```
# Import the necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Start coding here
workout_df = pd.read_csv("data/workout.csv")
ans1 = workout_df.sort_values(by="workout_worldwide", ascending=False)
ans1["month"] = pd.to_datetime(ans1['month'])
ans1['month'] = ans1['month'].dt.year
year_str = ans1['month'].iloc[0]
year_str = str(year_str)
```

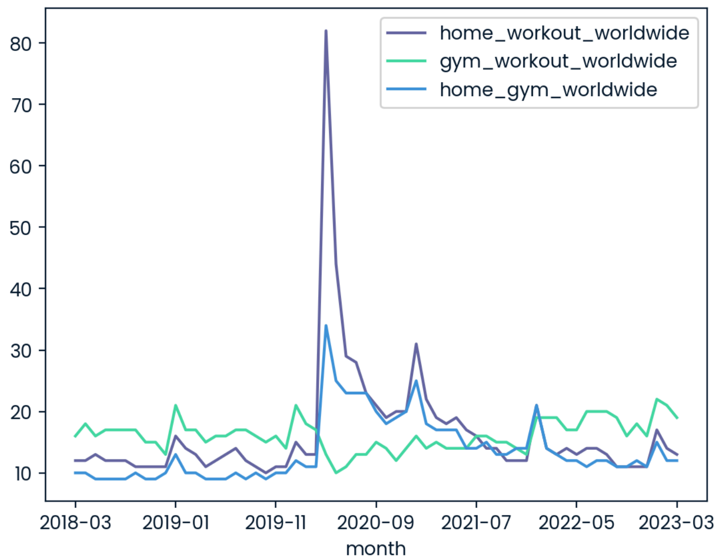
```
keywords_df = pd.read_csv("data/three_keywords.csv")
keywords_df
```

ind...	mo...	home_workout_worldwide	gym_workout_worldwide	home_gym_worldv
0	2018-03	12	16	
1	2018-04	12	18	
2	2018-05	13	16	
3	2018-06	12	17	
4	2018-07	12	17	
5	2018-08	12	17	
6	2018-09	11	17	
7	2018-10	11	15	
8	2018-11	11	15	
9	2018-12	11	13	
10	2019-01	16	21	
11	2019-02	14	17	
12	2019-03	13	17	
13	2019-04	11	15	
14	2019-05	12	16	
15	2019-06	13	16	

Rows: 61 Expand

```
keywords_df.plot('month',['home_workout_worldwide','gym_workout_worldwide','home_gym_worldwide'])
```

<AxesSubplot: xlabel='month'>



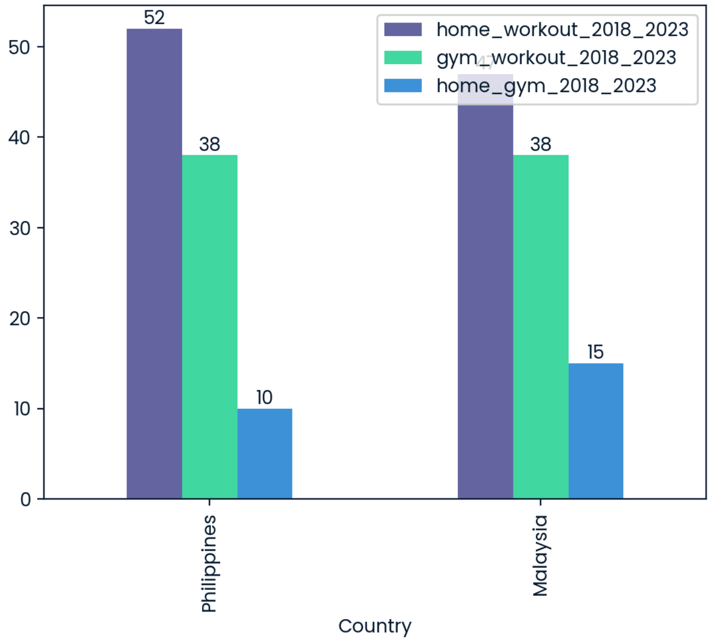

```
peak_covid = 'home_workout_worldwide'
current = 'gym_workout_worldwide'
```

```
workout_geo_df = pd.read_csv("data/workout_geo.csv")
workout_geo_df
ans3 = workout_geo_df.dropna()
ans3 = ans3.sort_values(by='workout_2018_2023', ascending=False)
top_country = ans3.iloc[0]['country']
```

```
# ans4
```

```
keywords_geo_df = pd.read_csv("data/three_keywords_geo.csv")
ans4 = keywords_geo_df
ans4 = ans4.dropna()
ans4 = ans4[ans4['Country'].isin(['Malaysia', 'Philippines'])]
# x_labels = ['Malaysia','Philippines']
ans4_plot = ans4.plot(kind='bar', x='Country')
for container in ans4_plot.containers:
    ans4_plot.bar_label(container)

plt.show()
```




```
home_workout_geo = 'Philippines'
```